

$$a \quad \frac{\sqrt{18}}{\sqrt{2}} = \sqrt{\frac{18}{2}} = \sqrt{9} = 3$$

$$b \quad \frac{\sqrt{5}}{\sqrt{2}} = \sqrt{\frac{5}{2}} = \sqrt{2\frac{1}{2}} \text{ or } \sqrt{2.5} \text{ or } \frac{1}{2}\sqrt{10}$$

$$c \quad \frac{\sqrt{16}}{\sqrt{7}} = \frac{\sqrt{16}}{\sqrt{7}} = \frac{4}{\sqrt{7}} = \frac{4 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}} = \frac{4\sqrt{7}}{7}$$

Example 8

Simplify $\frac{5\sqrt{7} \times 2\sqrt{3}}{\sqrt{45} \times \sqrt{21}}$.

$$\begin{aligned} \frac{5\sqrt{7} \times 2\sqrt{3}}{\sqrt{45} \times \sqrt{21}} &= \frac{5\sqrt{7} \times 2\sqrt{3}}{3\sqrt{5} \times \sqrt{3} \times \sqrt{7}} \\ &= \frac{5 \times 2}{3\sqrt{5}} \\ &= \frac{2\sqrt{5}}{3} \end{aligned}$$

Exercise 1f

Simplify the following by rationalising the denominators.

① $\frac{2}{\sqrt{2}}$

② $\frac{6}{\sqrt{2}}$

③ $\frac{10}{\sqrt{5}}$

④ $\frac{4}{\sqrt{8}}$

⑤ $\frac{21}{\sqrt{6}}$

⑥ $\frac{15}{\sqrt{3}}$

⑦ $\frac{\sqrt{4}}{\sqrt{5}}$

⑧ $\frac{\sqrt{9}}{\sqrt{7}}$

⑨ $\frac{21}{\sqrt{7}}$

⑩ $\frac{2\sqrt{3}}{\sqrt{6}}$

⑪ $\frac{8}{\sqrt{18}}$

⑫ $\frac{\sqrt{12}}{\sqrt{50}}$

⑬ $\frac{3\sqrt{2}}{\sqrt{10}}$

⑭ $\frac{30}{\sqrt{75}}$

⑮ $\frac{30}{\sqrt{72}}$

⑯ $\frac{\sqrt{18} \times \sqrt{20} \times \sqrt{24}}{\sqrt{8} \times \sqrt{30}}$

⑰ $\frac{\sqrt{3} \times \sqrt{18} \times \sqrt{39}}{\sqrt{24} \times \sqrt{26}}$

⑱ $\sqrt{3} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{27}}$

⑲ $2\sqrt{2} - \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{8}}$

⑳ $\frac{12}{\sqrt{24} - \sqrt{6}}$

1.4 Surds in brackets

Multiplication

When multiplying surds in brackets, use algebraic expansion:

$$(a + b)(c + d) = ac + ad + bc + bd$$

Example 9

Simplify the following.

a $(3\sqrt{5} + 2)(\sqrt{5} + 3)$

b $(3\sqrt{3} - 2\sqrt{2})(\sqrt{2} - 2\sqrt{3})$

c $(4\sqrt{3} + \sqrt{2})(4\sqrt{3} - \sqrt{2})$

d $(2\sqrt{2} + \sqrt{5})^2$

$$\begin{aligned} a \quad (3\sqrt{5} + 2)(\sqrt{5} + 3) &= (3\sqrt{5})\sqrt{5} + (3\sqrt{5})3 + 2(\sqrt{5}) + 2(3) \\ &= 15 + 9\sqrt{5} + 2\sqrt{5} + 6 \\ &= 21 + 11\sqrt{5} \end{aligned}$$

$$\begin{aligned} b \quad (3\sqrt{3} - 2\sqrt{2})(\sqrt{2} - 2\sqrt{3}) &= (3\sqrt{3})\sqrt{2} - (3\sqrt{3})(2\sqrt{3}) - (2\sqrt{2})\sqrt{2} \\ &\quad + (2\sqrt{2})(2\sqrt{3}) \\ &= 3\sqrt{6} - 18 - 4 + 4\sqrt{6} \\ &= 7\sqrt{6} - 22 \end{aligned}$$

$$\begin{aligned} c \quad (4\sqrt{3} + \sqrt{2})(4\sqrt{3} - \sqrt{2}) &= (4\sqrt{3})^2 - (\sqrt{2})^2 \\ &\quad \text{(difference of two squares)} \\ &= 48 - 2 \\ &= 46 \end{aligned}$$

$$\begin{aligned} d \quad (2\sqrt{2} + \sqrt{5})^2 &= (2\sqrt{2} + \sqrt{5})(2\sqrt{2} + \sqrt{5}) \\ &= 2\sqrt{2}(2\sqrt{2}) + 2\sqrt{2}(\sqrt{5}) + \sqrt{5}(2\sqrt{2}) \\ &\quad + \sqrt{5}(\sqrt{5}) \\ &= 8 + 2\sqrt{10} + 2\sqrt{10} + 5 \\ &= 13 + 4\sqrt{10} \end{aligned}$$

Exercise 1g

Simplify the following.

① $\sqrt{2}(\sqrt{2} + \sqrt{6})$

② $2\sqrt{5}(3\sqrt{5} - 2\sqrt{2})$

③ $(\sqrt{6} - \sqrt{5})(\sqrt{8} + \sqrt{15})$

④ $(\sqrt{5} + \sqrt{15})(2\sqrt{3} - 1)$

- ⑤ $(3\sqrt{2} + \sqrt{5})(3\sqrt{2} - \sqrt{5})$
 ⑥ $(4\sqrt{3} + 2\sqrt{5})(4\sqrt{3} - 2\sqrt{5})$
 ⑦ $(\sqrt{7} - \sqrt{3})(2\sqrt{7} + \sqrt{3})$
 ⑧ $(3\sqrt{2} - \sqrt{6})(2\sqrt{3} + \sqrt{2})$
 ⑨ $(2\sqrt{2} + \sqrt{5})(\sqrt{5} - 2\sqrt{2})$
 ⑩ $(\sqrt{6} + 2\sqrt{3})^2$
 ⑪ $(\sqrt{5} - 2\sqrt{10})^2$
 ⑫ $(1 + 3\sqrt{3})^2$
 ⑬ $\left(\sqrt{2} + \frac{1}{\sqrt{2}}\right)^2$
 ⑭ $(3\sqrt{2} - 2\sqrt{3})(5\sqrt{2} - \sqrt{3})$
 ⑮ $(3\sqrt{2} - \sqrt{5})^2$

Conjugate of a binomial surd

An expression may contain two terms, for example $6 - \sqrt{5}$, $3\sqrt{2} + \sqrt{3}$, $2\sqrt{3} + 4$. If one or both of the terms contains a surd we call this a **binomial surd** expression. To rationalise a binomial surd, we use the idea of **the difference of two squares**.

For example, to rationalise $\sqrt{a} + b$ we multiply it by $\sqrt{a} - b$ to get $(\sqrt{a})^2 - b^2$, which is a rational number. $\sqrt{a} + b$ and $\sqrt{a} - b$ are said to be **conjugates** of one another.

Example 10

Evaluate the following.

a $(2\sqrt{3} - 4)(2\sqrt{3} + 4)$

b $(5 + 3\sqrt{2})(5 - 3\sqrt{2})$

$$\begin{aligned} \text{a } (2\sqrt{3} - 4)(2\sqrt{3} + 4) &= (2\sqrt{3})^2 - 4^2 \\ &= 12 - 16 \\ &= -4 \end{aligned}$$

$$\begin{aligned} \text{b } (5 + 3\sqrt{2})(5 - 3\sqrt{2}) &= 5^2 - (3\sqrt{2})^2 \\ &= 25 - 18 \\ &= 7 \end{aligned}$$

If the denominator of a fraction is a binomial surd, use the conjugate of the denominator to rationalise it.

Example 11

Simplify the following by rationalising the denominators.

a $\frac{2}{3\sqrt{5} + 4}$ b $\frac{6}{2\sqrt{2} - 1}$ c $\frac{2\sqrt{3} + 2}{2\sqrt{3} - 2}$

$$\begin{aligned} \text{a } \frac{2}{3\sqrt{5} + 4} &= \frac{2(3\sqrt{5} - 4)}{(3\sqrt{5} + 4)(3\sqrt{5} - 4)} \\ &= \frac{6\sqrt{5} - 8}{(3\sqrt{5})^2 - 4^2} \\ &= \frac{6\sqrt{5} - 8}{45 - 16} \\ &= \frac{6\sqrt{5} - 8}{29} \end{aligned}$$

$$\begin{aligned} \text{b } \frac{6}{2\sqrt{2} - 1} &= \frac{6(2\sqrt{2} + 1)}{(2\sqrt{2} - 1)(2\sqrt{2} + 1)} \\ &= \frac{12\sqrt{2} + 6}{(2\sqrt{2})^2 - 1^2} \\ &= \frac{12\sqrt{2} + 6}{7} \end{aligned}$$

$$\begin{aligned} \text{c } \frac{2\sqrt{3} + 2}{2\sqrt{3} - 2} &= \frac{(2\sqrt{3} + 2)(2\sqrt{3} + 2)}{(2\sqrt{3} - 2)(2\sqrt{3} + 2)} \\ &= \frac{12 + 4\sqrt{3} + 4\sqrt{3} + 4}{(2\sqrt{3})^2 - 2^2} \\ &= \frac{16 + 8\sqrt{3}}{12 - 4} \\ &= \frac{8(2 + \sqrt{3})}{8} \\ &= 2 + \sqrt{3} \end{aligned}$$

Exercise 1h

Simplify the following by rationalising the denominators.

① $\frac{1}{2 - \sqrt{3}}$

② $\frac{4}{3 + \sqrt{7}}$

③ $\frac{\sqrt{5}}{\sqrt{15} - \sqrt{10}}$

④ $\frac{3\sqrt{7}}{5 - \sqrt{7}}$

⑤ $\frac{\sqrt{6} + \sqrt{2}}{\sqrt{6} - \sqrt{2}}$

⑥ $\frac{2\sqrt{3} - \sqrt{5}}{2\sqrt{3} + \sqrt{5}}$

⑦ $\frac{\sqrt{3} + \sqrt{5}}{\sqrt{5} - 2}$

⑧ $\frac{\sqrt{3} + \sqrt{6}}{3\sqrt{2} - 3}$

⑨ $\frac{\sqrt{5} + 3}{4 - \sqrt{10}}$

⑩ $\frac{2\sqrt{5} + \sqrt{3}}{5\sqrt{3} - 3\sqrt{5}}$

$$(11) \frac{\sqrt{7} + 3\sqrt{3}}{2\sqrt{7} + \sqrt{3}}$$

$$(13) \frac{\sqrt{7} - 3\sqrt{2}}{2\sqrt{7} - \sqrt{3}}$$

$$(15) \left(\frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}} \right)^2$$

$$(12) \frac{3 - 2\sqrt{2}}{3 + 2\sqrt{2}}$$

$$(14) \frac{2\sqrt{6} - \sqrt{3}}{2\sqrt{3} - 1}$$

1.5 Evaluating expressions with surds

When evaluating a fraction containing a surd, it is advisable to rationalise its denominator.

Example 12

Given that $\sqrt{3} = 1.732$, evaluate $\frac{2}{\sqrt{3}}$ to 2 d.p.

$$\begin{aligned} \frac{2}{\sqrt{3}} &= \frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{2\sqrt{3}}{3} \\ &= \frac{2 \times 1.732}{3} = \frac{3.464}{3} \\ &= 1.154... \\ &= 1.15 \text{ to 2 d.p.} \end{aligned}$$

Notice the advantage of rationalising the denominator. Without this process:

$$\begin{aligned} \frac{2}{\sqrt{3}} &= \frac{2}{1.732} = 1.154... \\ &= 1.15 \text{ to 2 d.p.} \end{aligned}$$

The last step involves division by 1.732. This would normally require the use of tables or a calculator.

Other expressions can sometimes be evaluated without taking roots at all. See Example 13.

Example 13

Without using tables or a calculator, evaluate $\sqrt{\frac{2}{3}}(\sqrt{0.54} + \sqrt{6})$.

$$\begin{aligned} \sqrt{\frac{2}{3}}(\sqrt{0.54} + \sqrt{6}) &= \sqrt{\frac{2}{3}} \times \sqrt{0.54} + \sqrt{\frac{2}{3}} \times \sqrt{6} \\ &= \sqrt{\frac{2}{3} \times 0.54} + \sqrt{\frac{2}{3} \times 6} \\ &= \sqrt{0.36} + \sqrt{4} \\ &= 0.6 + 2 \\ &= 2.6 \end{aligned}$$

Notice that when evaluating expressions of this kind, it is usual to take the positive square root only.

Exercise 1i

Do not use tables or calculators in this exercise.

(1) Given that $\sqrt{2} = 1.414$ and $\sqrt{3} = 1.732$, evaluate the following correct to 3 s.f.

a $\frac{1}{\sqrt{3}}$

b $\frac{1}{\sqrt{2}}$

c $\frac{2}{\sqrt{2}}$

d $\frac{3}{\sqrt{3}}$

e $\frac{10}{\sqrt{2}}$

f $\frac{6}{\sqrt{3}}$

(2) If $\sqrt{5} = 2.236$, evaluate $\sqrt{\frac{7 \times 23 - 1}{8}}$ correct to 2 d.p.

(3) Evaluate $\sqrt{3^2 \times 0.25 \times 8}$ correct to 2 d.p., given that $\sqrt{2} = 1.414$.

(4) $\sqrt{1.225} = 1.107$, $\sqrt{12.25} = 3.5$ and $\sqrt{100} = 10$. Evaluate $\sqrt{1225}$. [WAEC]

(5) Evaluate the following.

a $\frac{\sqrt{8}}{\sqrt{50} - \sqrt{2}}$

b $(\sqrt{19} + \sqrt{11})(\sqrt{19} - \sqrt{11})$

c $(5\sqrt{2.5} - \sqrt{10})\sqrt{0.4}$

d $(\sqrt{0.6} - \sqrt{15})^2$ [WAEC]

1.6 Trigonometric ratios of 30° , 60° and 45°

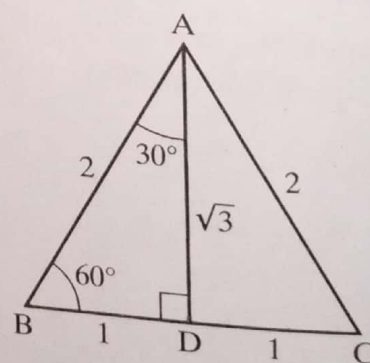


Fig. 1.1

In Fig. 1.1, ABC is an equilateral triangle of side 2 units.

The altitude $\overline{AD}$ is $\sqrt{3}$ units long (Pythagoras). It follows that:

$$\sin 30^\circ = \frac{1}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}, \tan 30^\circ = \frac{1}{\sqrt{3}},$$

and

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 60^\circ = \frac{1}{2}, \tan 60^\circ = \sqrt{3}.$$

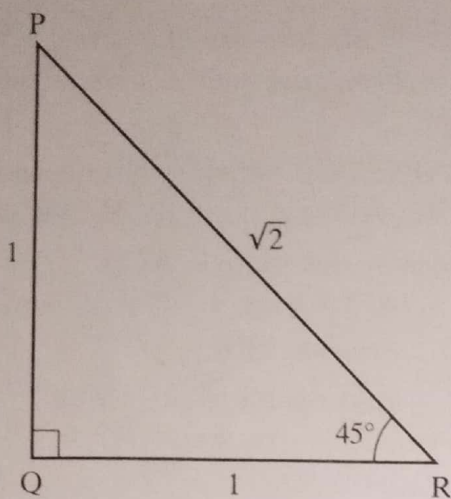


Fig. 1.2

In Fig. 1.2, PQR is an isosceles right-angled triangle with two sides of 1 unit. Its hypotenuse $\overline{PR}$ is $\sqrt{2}$ units long (Pythagoras).

It follows that:

$$\sin 45^\circ = \frac{1}{\sqrt{2}}, \cos 45^\circ = \frac{1}{\sqrt{2}}, \tan 45^\circ = 1.$$

Example 14

In Fig. 1.3, if $|AB| = 6$ cm, calculate x and y . (Leave the answers in surd form with rational denominators.)

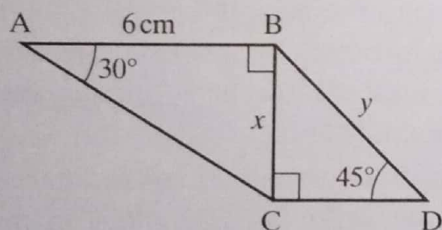


Fig. 1.3

In $\triangle ABC$,

$$\tan 30^\circ = \frac{x}{6}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{x}{6}$$

$$\begin{aligned} \Rightarrow x &= \frac{6}{\sqrt{3}} \text{ cm} \\ &= \frac{6\sqrt{3}}{3} \text{ cm} \\ &= 2\sqrt{3} \text{ cm} \end{aligned}$$

In $\triangle BCD$,

$$\tan 45^\circ = \frac{x}{y}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{2\sqrt{3}}{y}$$

$$\begin{aligned} \Rightarrow y &= 2\sqrt{3} \times \sqrt{2} \text{ cm} \\ &= 2\sqrt{6} \text{ cm} \end{aligned}$$

Exercise 1j

In each part of Fig. 1.4, calculate the lengths marked x and y . All dimensions are in centimetres. Give answers in surd form with rational denominators.

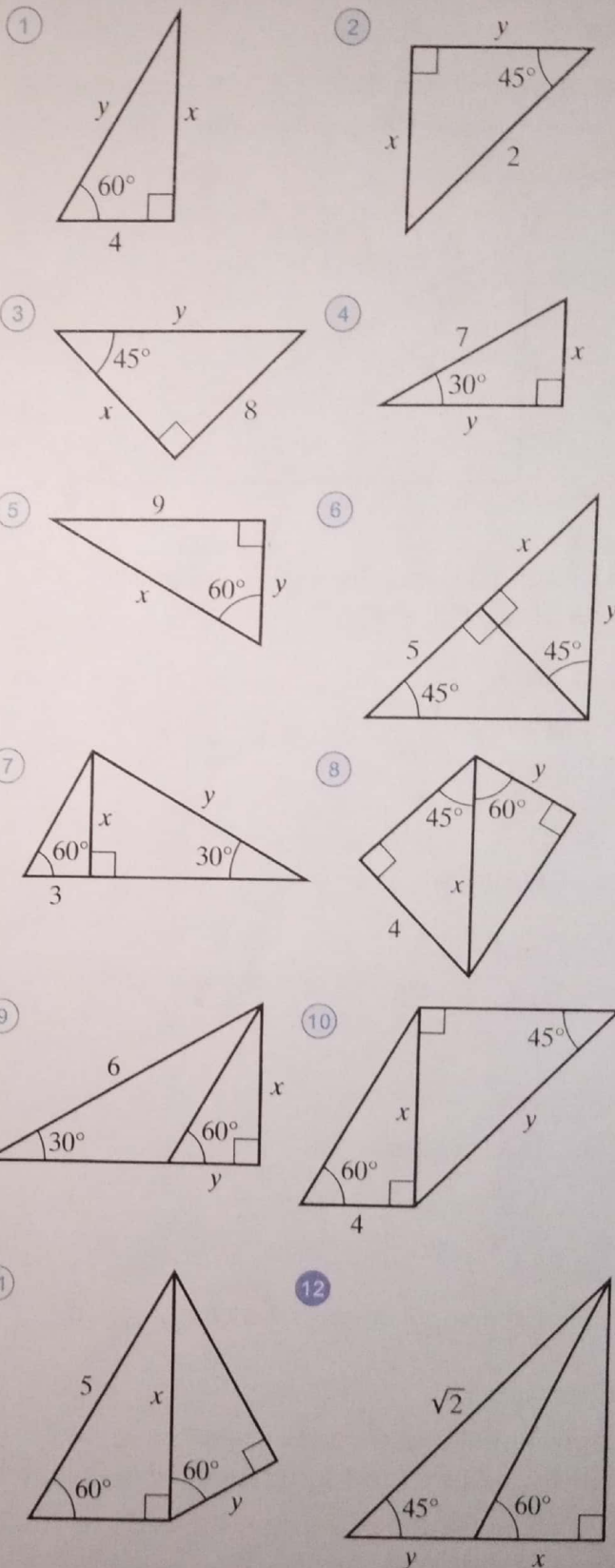


Fig. 1.4

Example 15

From the top of a tower 80 m high, two boats are seen in a direction due south. The angles of depression of the boats from the top of the tower are 45° and 30° . Find the distance between the boats.

First, draw a sketch. In Fig. 1.5, $\overline{HT}$ represents the tower; A and B are the positions of the boats.

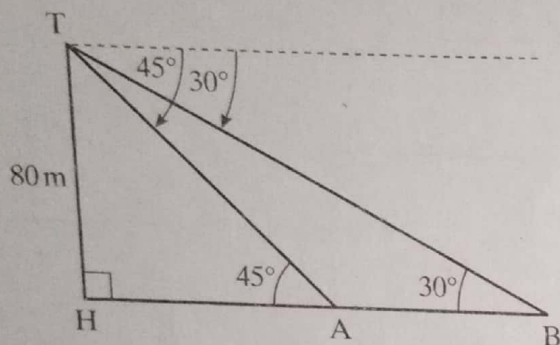


Fig. 1.5

$|AB|$ is the distance between the boats.

$$|AB| = |HB| - |HA|$$

In $\triangle THB$,

$$\tan 30^\circ = \frac{80}{|HB|}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{80}{|HB|}$$

$$\Rightarrow |HB| = 80\sqrt{3} \text{ m}$$

In $\triangle THA$,

$$\tan 45^\circ = \frac{80}{|HA|}$$

$$\Rightarrow 1 = \frac{80}{|HA|}$$

$$\Rightarrow |HA| = 80 \text{ m}$$

$$\begin{aligned} \Rightarrow |AB| &= 80\sqrt{3} - 80 \text{ m} \\ &= 80(\sqrt{3} - 1) \text{ m} \end{aligned}$$

Notice the importance of drawing a sketch.

Exercise 1k

Draw a sketch in each question. Leave the answers in surd form with rational denominators.

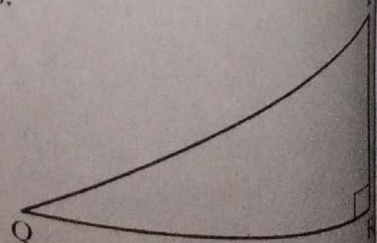
- From the top of a tower, the angle of depression of a boat is 30° . If the tower is 20 m high, how far is the boat from the foot of the tower? [WAEC]

- The angle of elevation of X from Y is 30° . If $|XY| = 40 \text{ m}$, how high is X above the level of Y? [WAEC]
- In $\triangle ABC$, $|AB| = 6 \text{ cm}$, $\hat{A} = 90^\circ$ and $\hat{B} = 30^\circ$, $\overline{AD}$ is an altitude. Find $|CD|$.
- In the isosceles triangle ACB, $|AB| = |AC| = 4 \text{ cm}$, $\hat{BAC} = 30^\circ$ and $\overline{CN}$ is an altitude. Find $|BN|$.
- XYZ is an isosceles triangle with $|XY| = |XZ| = 6 \text{ cm}$ and $\hat{YXZ} = 120^\circ$. Calculate the length of $\overline{YZ}$. [WAEC]
- A cone has a circular base, a perpendicular height of 21 cm, and a semi-vertical angle of 30° . Calculate the slant height of the cone. Find the area of its base. (Take π to be $\frac{22}{7}$.)
- When the angle of elevation of the sun is 30° the shadow of a vertical tower is 20 m longer than when the elevation of the sun is 60° . Find the height of the tower.
- Two huts and a radio mast are on level ground such that one hut is due east of the mast and the other is due west of it. From the top of the mast the angles of depression of the huts are 60° and 45° , respectively. If the mast is 150 m high, find the distance between the huts.
- A and B are two points on level ground, both due south of a flagpole. The angle of elevation of the top of the flagpole is 60° from A and 45° from B. If A is 20 m from the foot of the flagpole, find $|AB|$.
- The top of a building 24 m high is observed from the top and from the bottom of a vertical tree. The angles of elevation are found to be 45° and 60° , respectively. Find the height of the tree.

1.7 Pythagoras' theorem

In a right-angled triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides.

Fig. 1.6



In Fig. 1.6,

$$PQ^2 = PR^2 + QR^2$$

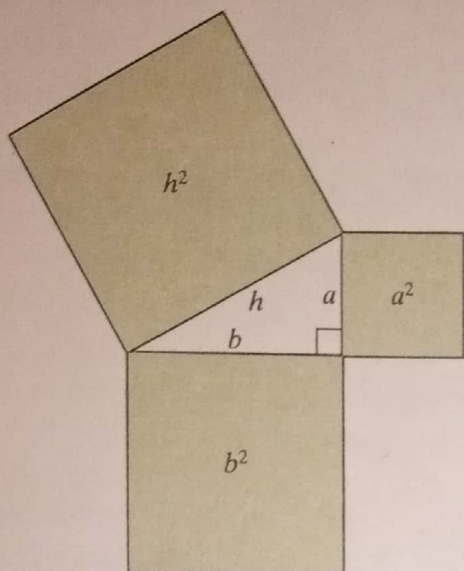


Fig. 1.7

In Fig. 1.7, the triangle has sides of lengths h , a and b units.

$$h^2 = a^2 + b^2$$

also:

$$a^2 = h^2 - b^2$$

$$b^2 = h^2 - a^2$$

Example 16

In Fig. 1.8, $|AC| = 11$ cm, $|BC| = 4$ cm and $|CD| = 10$ cm. Find $|AD|$.

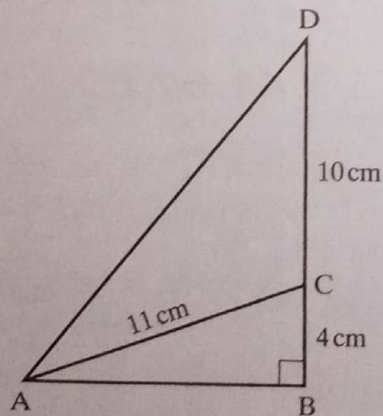


Fig. 1.8

In $\triangle ABC$,

$$\begin{aligned} AB^2 &= AC^2 - BC^2 \\ &= 11^2 - 4^2 \\ &= 121 - 16 = 105 \end{aligned}$$

In $\triangle ABD$,

$$\begin{aligned} AD^2 &= AB^2 + BD^2 \\ &= 105 + 14^2 \\ &= 105 + 196 = 301 \\ |AD| &= \sqrt{301} \text{ cm} \end{aligned}$$

Exercise 11

In this exercise, leave all answers in surd form, simplified as far as possible.

- ① The dimensions in Fig. 1.9 are all centimetres. In each case, x is a whole number. Find the value of x in each part.

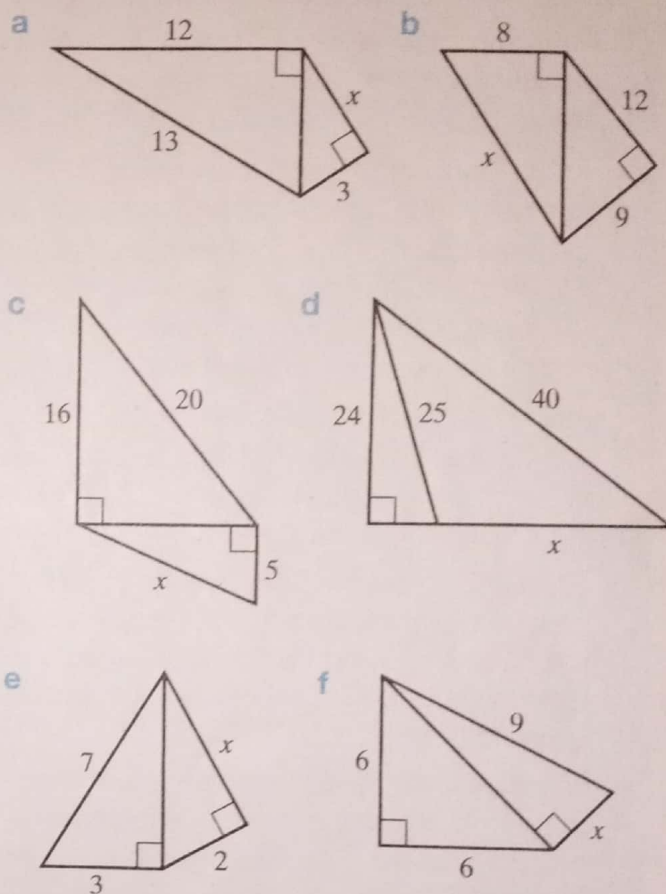


Fig. 1.9

- ② The dimensions in Fig. 1.10 are all centimetres. Find the value of x in each part.

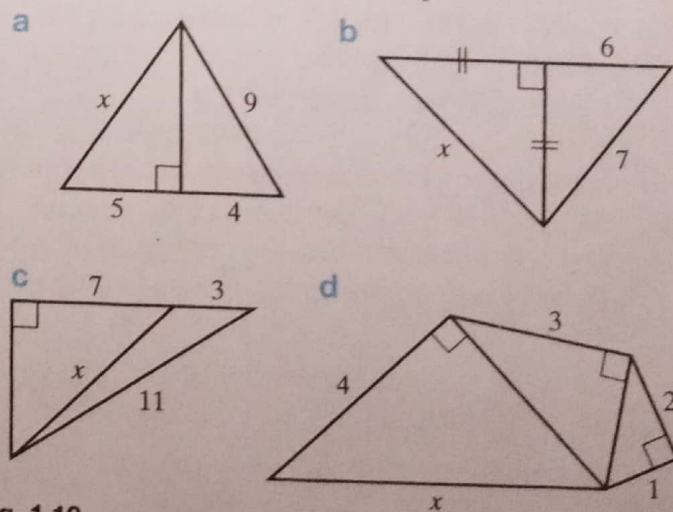


Fig. 1.10

- ③ Find the perpendicular height of a cone which has a slant height of 10 cm and a circular base of diameter 10 cm.

- 4 A vertical radio mast is supported by wires attached to its top and to points on the level ground 9 m from its base. If each wire is 31 m long, find the height of the mast.
- 5 Village B is 7 km due south of village A and village C is 9 km due south of B. Village D is 8 km from A and is due east of B. Find the distance of D from: **a** B **b** C.

SUMMARY

- 1 A *surd* is an irrational or *non-rational* number of the form $a\sqrt{n}$, where n is not a perfect square. Thus, for example, $\sqrt{2}$, $2\sqrt{3}$, $\sqrt{10}$ and $-2\sqrt{5}$ are surds but $\sqrt{4}$, $2\sqrt{9}$, $\sqrt{100}$ are not. *Irrational* numbers cannot be expressed as ratios in the form $\frac{a}{b}$.
- 2 Like surds have the same numbers under the square root sign: $2\sqrt{5}$ and $6\sqrt{5}$ are like surds. We can add and subtract like surds. For example, $10\sqrt{m} - 7\sqrt{m} - \sqrt{m} = 4\sqrt{m}$.
- 3 To *simplify a surd* usually means to find an equivalent surd in which the number under the square root sign is as small as possible. Examples 1 and 3–5 show various forms of simplification.
- 4 To *rationalise the denominator* means to make the denominator of an expression a rational number. See Examples 6–8.
- 5 Clear brackets with surds by following the rules of algebra. See Example 9.
- 6 A *binomial surd* contains two terms, e.g. $(3 + \sqrt{2})$ and $(5\sqrt{3} - 3\sqrt{5})$ are binomial surds. One binomial surd is the *conjugate* of another, if they differ only in the sign connecting the two terms, e.g. $(3 - \sqrt{2})$ and $(5\sqrt{3} + 3\sqrt{5})$ are the respective conjugates of the binomials in the previous sentence. If a fraction has a binomial surd in its denominator, use the conjugate of the denominator to rationalise it. See Examples 10 and 11.
- 7 Some of the trigonometric ratios of special angles can be expressed in surd form.

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 60^\circ = \sqrt{3}$$

Challenge Box

Some sums

	4	3	2
+	6	5	7
	1	0	8
			9

Ten cards, numbered 0 to 9, are arranged to make a correct addition sum.

Find other ways of arranging the ten cards so that two 3-figure numbers add correctly to give a 4-figure number.

Challenge Box

Primes

Here is a set of six prime numbers which use the digits 1 to 9 exactly once.

2, 3, 5, 41, 67, 89

Find more ways of doing this (i.e. make other sets of six prime numbers using the digits 1 to 9 once only)

2

Numerical processes (2) Theory of logarithms

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Express statements given in index form (such as $81 = 3^4$) as an equivalent logarithm statement ($\log_3 81 = 4$)
- 2 Evaluate expressions given in logarithmic form
- 3 Note the equivalence between the laws of indices and the laws of logarithms
- 4 Recall and use the laws of logarithms to simplify and/or evaluate given expressions without the use of logarithm tables
- 5 Use logarithm tables for the purposes of calculation.

Teaching and learning materials

Teacher: Chart demonstrating laws of logarithms; logarithm tables (as used by Examinations Boards)

Student: Logarithm and antilogarithm tables (see pages 272–273 of this book)

2.1 Logarithms and indices

Books 1 and 2 of *New General Mathematics for Senior Secondary Schools* showed that the statements $100 = 10^2$ and $\log 100 = 2$ are two ways of writing the same thing. The second statement in full is $\log_{10} 100 = 2$, i.e. 'the logarithm to the base 10 of 100 is 2'.

Logarithms can be in bases other than 10. For example, since $8 = 2^3$, $\log_2 8 = 3$ or, in words, the logarithm to the base 2 of 8 is 3.

In general, if $N = a^x$, then $\log_a N = x$.

The logarithm to a given base of a number is the power to which the base must be raised to make the number.

Example 1

Evaluate: a $\log_2 32$ b $\log_9 27$ c $\log_5 0.04$

a Let $N = \log_2 32$
then $2^N = 32 = 2^5$
hence $N = 5$ (equating indices)

b Let $x = \log_9 27$
then $9^x = 27$
i.e. $(3^2)^x = 3^3$
 $3^{2x} = 3^3$
hence $2x = 3$ (equating indices)
and $x = 1\frac{1}{2}$

c Let $r = \log_5 0.04$
then $5^r = 0.04 = \frac{1}{25} = 5^{-2}$
hence $r = -2$ (equating indices)

Exercise 2a

Evaluate the following logarithms.

- | | |
|-------------------------|------------------------|
| ① $\log_2 4$ | ② $\log_{10} 1\,000$ |
| ③ $\log_5 25$ | ④ $\log_3 81$ |
| ⑤ $\log_{12} 144$ | ⑥ $\log_6 216$ |
| ⑦ $\log_7 \frac{1}{7}$ | ⑧ $\log_4 64$ |
| ⑨ $\log_4 8$ | ⑩ $\log_8 4$ |
| ⑪ $\log_{25} 0.2$ | ⑫ $\log_{49} 7$ |
| ⑬ $\log_{1000} 10$ | ⑭ $\log_4 \frac{1}{8}$ |
| ⑮ $\log_{16} 0.25$ | ⑯ $\log_{100} 0.001$ |
| ⑰ $\log_9 \frac{1}{27}$ | ⑱ $\log_8 0.0625$ |
| ⑲ $\log_{1.2} 1.728$ | ⑳ $\log_{0.2} 25$ |

2.2 Laws of logarithms

The three fundamental laws of indices can be stated in their equivalent logarithmic form:

$$x^a \times x^b = x^{a+b}$$

$$\log(MN) = \log M + \log N$$

$$x^a \div x^b = x^{a-b}$$

$$\log \left(\frac{M}{N} \right) = \log M - \log N$$

$$(x^a)^b = x^{ab}$$

$$\log(M^p) = p \log M$$

Example 2

Prove that $\log(MN) = \log M + \log N$.

Let $\log_x M = a$ and $\log_x N = b$, where x is any base.

Then $M = x^a$ and $N = x^b$

$$\begin{aligned} MN &= x^a \times x^b \\ &= x^{a+b} \end{aligned}$$

$$\log_x(MN) = a + b = \log_x M + \log_x N$$

Notice in Example 2 that since x can take any value, the law

$$\log(MN) = \log M + \log N$$

is true for *any* base. This is true for all of the laws of logarithms.

Example 3

Given that $\log 2 = 0.301\,03$ and $\log 3 = 0.477\,12$, calculate, without using any tables

$$a \log 5 \quad b \log 6 \quad c \log 0.9$$

[WAEC]

$$\begin{aligned} a \log 5 &= \log \frac{10}{2} \\ &= \log 10 - \log 2 \\ &= 1 - 0.301\,03 \\ &= 0.698\,97 \end{aligned}$$

$$\begin{aligned} b \log 6 &= \log(2 \times 3) \\ &= \log 2 + \log 3 \\ &= 0.301\,03 + 0.477\,12 \\ &= 0.778\,15 \end{aligned}$$

$$\begin{aligned} c \log 0.9 &= \log \frac{9}{10} \\ &= \log 9 - \log 10 \\ &= \log 3^2 - 1 \\ &= 2 \log 3 - 1 \\ &= 2 \times 0.477\,12 - 1 \\ &= 0.954\,24 - 1 \\ &= -0.045\,76 \end{aligned}$$

Example 4

Simplify the following as far as possible.

$$a \log 8 + \log 5$$

$$b \quad 2 - \log 4$$

$$c \log 9 \div \log 3$$

$$d \quad \frac{2}{5} \log 32$$

$$\begin{aligned} a \log 8 + \log 5 &= \log(8 \times 5) \\ &= \log 40 \end{aligned}$$

$$\begin{aligned} b \quad 2 - \log 4 &= \log 100 - \log 4 \\ &= \log \frac{100}{4} \\ &= \log 25 \end{aligned}$$

$$\begin{aligned} c \log 9 \div \log 3 &= \frac{\log 9}{\log 3} = \frac{\log 3^2}{\log 3} = \frac{2 \log 3}{\log 3} \\ &= 2 \end{aligned}$$

$$\begin{aligned} d \quad \frac{2}{5} \log 32 &= \log 32^{\frac{2}{5}} = \log (2^5)^{\frac{2}{5}} = 2 \log 2 \\ [or] &= \log (\sqrt[5]{32})^2 = \log 2^2 \\ &= \log 4 \end{aligned}$$

Example 5

Evaluate $\log_3 6.84$ to 2 decimal places.

$$\text{Let } \log_3 6.84 = x$$

$$\text{then } 3^x = 6.84$$

$$\Leftrightarrow \log_{10}(3^x) = \log_{10} 6.84$$

$$\Leftrightarrow x \log_{10} 3 = \log_{10} 6.84$$

working:

$$\begin{aligned} \Leftrightarrow x &= \frac{\log_{10} 6.84}{\log_{10} 3} \\ &= \frac{0.8351}{0.4771} \\ &= 1.75 \text{ to 2 d.p.} \end{aligned}$$

No	Log
0.8351	1.9218
0.4771	-1.6786
1.751	0.2432

Exercise 2b

Unless otherwise stated, assume that all the logarithms are to base 10.

① Given that $\log 2 = 0.301\,0$, $\log 3 = 0.477\,1$ and $\log 7 = 0.845\,1$, evaluate the following.

$$a \log 5$$

$$b \log 8$$

$$c \log 49$$

$$d \log 14$$

$$e \log 35$$

$$f \log 42$$

② Given that $\log 2 = 0.301\,01$, evaluate $\log 16$ without using tables.

[WAEC]

③ Express the following as logarithms of single numbers.

$$a \log 3 + \log 4$$

$$b \log 15 - \log 3$$

$$c \quad 3 \log 5$$

$$d \quad \frac{1}{2} \log 25$$

- e $\log 81 - \log 3$ f $1 + \log 3$
 g $1 - \log 5$ h $2\log 3 + \log 6$
 i $2 - 2\log 5$ j $\frac{3}{4}\log 16$

④ Evaluate, without any tables,
 $3\log 2 + \log 20 - \log 1.6$. [WAEC]

⑤ Solve the following for x .

- a $\log_{10} x = 3$ b $\log_{10} x = -2$
 c $\log_x 81 = 4$ d $2\log_x (3\frac{3}{8}) = 6$

⑥ Simplify the following without using tables.

- a $\frac{\log 4}{\log 2}$ b $\frac{\log 16}{\log 8}$ c $\frac{\log 243}{\log 3}$
 d $\frac{\log \sqrt{5}}{\log 5}$ e $\frac{\log \sqrt{2}}{\log 9}$ f $\frac{\log 0.2}{\log 25}$

⑦ Simplify the following.

- a $\log 8 - \log 4$ b $\log 8 \div \log 4$
 c $\log 8 + \log 4$ d $\frac{\log 8 - \log 4}{\log 4 - \log 2}$

⑧ Express the following in index form.

- a $\log x = b$ b $\log x + \log y = 1$
 c $\log x + 1 = 0$ d $\log x + 2\log y = 3$
 e $\log x - \log y = \log z$

⑨ If $\log x - \log(2x - 1) = 1$, find x . [WAEC]

⑩ Use the method of Example 2 to prove:

- a $\log\left(\frac{M}{N}\right) = \log M - \log N$
 b $\log(M^p) = p \log M$

⑪ Use logarithm tables to evaluate the following, giving answers to two decimal places.

- a $\log_2 6$ b $\log_7 6$
 c $\log_5 12.64$ d $\log_{12} 0.8621$

⑫ Solve the following equations.

- a $(2^x)^2 - 3 \times 2^x + 2 = 0$ (Hint: let $y = 2^x$)
 b $2^{2x} - 10 \times 2^x + 16 = 0$

SUMMARY

1 The Laws of indices are:

$$x^a \times x^b = x^{a+b}$$

$$x^a \div x^b = x^{a-b}$$

$$x^0 = 1 \text{ (for all } x \neq 0\text{)}$$

$$(x^a)^b = x^{ab}$$

$$x^{-a} = \frac{1}{x^a}$$

$$x^{\frac{1}{a}} = \sqrt[a]{x}$$

$$x^{\frac{a}{b}} = \sqrt[b]{x^a} \text{ or } (\sqrt[b]{x})^a$$

These laws are true for all values of a , b and $x \neq 0$.

2 Logarithm is another word for power.
 For example:

powers	logarithms
$1000 = 10^3$	$\log 1000 = 3$
$100 = 10^2$	$\log 100 = 2$
$10 = 10^1$	$\log 10 = 1$
$1 = 10^0$	$\log 1 = 0$
$0.1 = 10^{-1}$	$\log 0.1 = -1$
$0.01 = 10^{-2}$	$\log 0.01 = -2$

Generally, if $x = a^b$, then $\log_a x = b$

3 The Laws of logarithms are:

$$\log(MN) = \log M + \log N$$

$$\log\left(\frac{M}{N}\right) = \log M - \log N$$

$$\log(M)^p = p \log M$$

2.3 Calculation using logarithm tables

Pages 139–140 of Chapter 14 contain a revision of how to use logarithm tables for calculations. Turn to those pages, study the examples 3–10 and do some of the exercises.

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Solve quadratic equations using factorisation, completing the square, the quadratic formula, and graphical methods
- 2 Transform word problems into linear equations that may be solved algebraically
- 3 Transform word problems into quadratic equations that may be solved algebraically
- 4 Use algebraic and/or graphical methods to solve simultaneous linear and quadratic equations.

Teaching and learning materials

Teacher: Graph boards; aids for drawing curves (wire, twine)

Students: Graph books; aids for drawing curves (wire, twine)

3.1 Solution of quadratic equations

Books 1 and 2 of *New General Mathematics for Senior Secondary Schools* showed the following methods for solving quadratic equations:

- 1 factorisation
- 2 completing the square
- 3 use of the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- 4 drawing graphs.

Examples 1–4 demonstrate these methods.

The method used in Example 1 involves factorising the quadratic expression in the given equation. Each of the resulting linear expressions is equated to zero and the solutions are found.

In general, if $P \times Q = 0$, then $P = 0$ or $Q = 0$ or both P and $Q = 0$.

Example 1

Solve the equation $2x^2 + 9x = 5$.

$$2x^2 + 9x = 5$$

$$\Rightarrow 2x^2 + 9x - 5 = 0$$

$$\Rightarrow (2x - 1)(x + 5) = 0$$

$$\text{Either } 2x - 1 = 0 \text{ or } x + 5 = 0$$

$$\Rightarrow 2x = 1 \quad \text{or} \quad x = -5$$

$$\Rightarrow x = \frac{1}{2} \quad \text{or} \quad x = -5$$

Sometimes a quadratic expression cannot be factorised. In such cases, other methods can be used. Example 2 shows the method of completing the square.

Example 2

Solve the equation $2x^2 - x - 8 = 0$, giving the solutions correct to two decimal places.

First, arrange the equation so that the coefficient of x^2 is 1 and the constant term is on the RHS.

$$x^2 - \frac{1}{2}x = 4$$

Make the LHS a perfect square by adding $\left(-\frac{1}{4}\right)^2$ to both sides of the equation.

$$x^2 - \frac{1}{2}x + \left(-\frac{1}{4}\right)^2 = 4 + \left(-\frac{1}{4}\right)^2$$

$$= 4\frac{1}{16}$$

$$\Rightarrow \left(x - \frac{1}{4}\right)^2 = 4\frac{1}{16}$$

$$\Rightarrow x - \frac{1}{4} = \pm \sqrt{4\frac{1}{16}}$$

$$= \pm \frac{8.062}{4}$$

$$\Rightarrow x = \frac{1}{4} + \frac{8.062}{4} \text{ or } \frac{1}{4} - \frac{8.062}{4}$$

$$\Rightarrow x \approx 2.27 \text{ or } -1.77$$

Example 3 shows how to use the general formula to solve a quadratic equation. *Note:* If a quadratic equation can be factorised, do not use the formula.